

Pathfinder

Smart Cane
for the Visually Impaired

Sponsored By:



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Cyber-Physical Systems Engineering

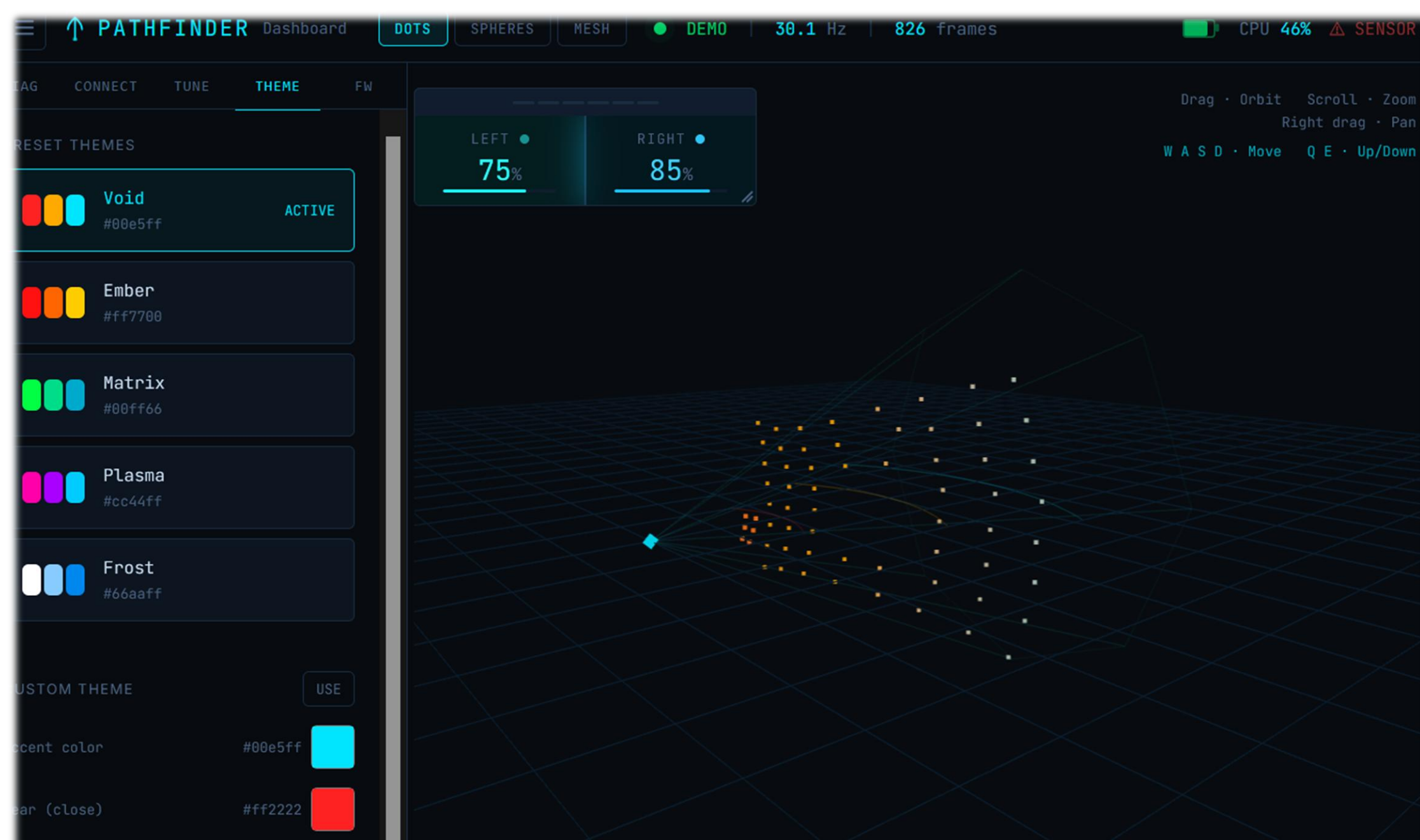
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The Universities
AT SHADY GROVE

THE PROBLEM

- **240 million** people with visual impairments worldwide
- **20 million** use white canes
- Overhead obstacle detection is difficult
- Limited assistance for independent travel



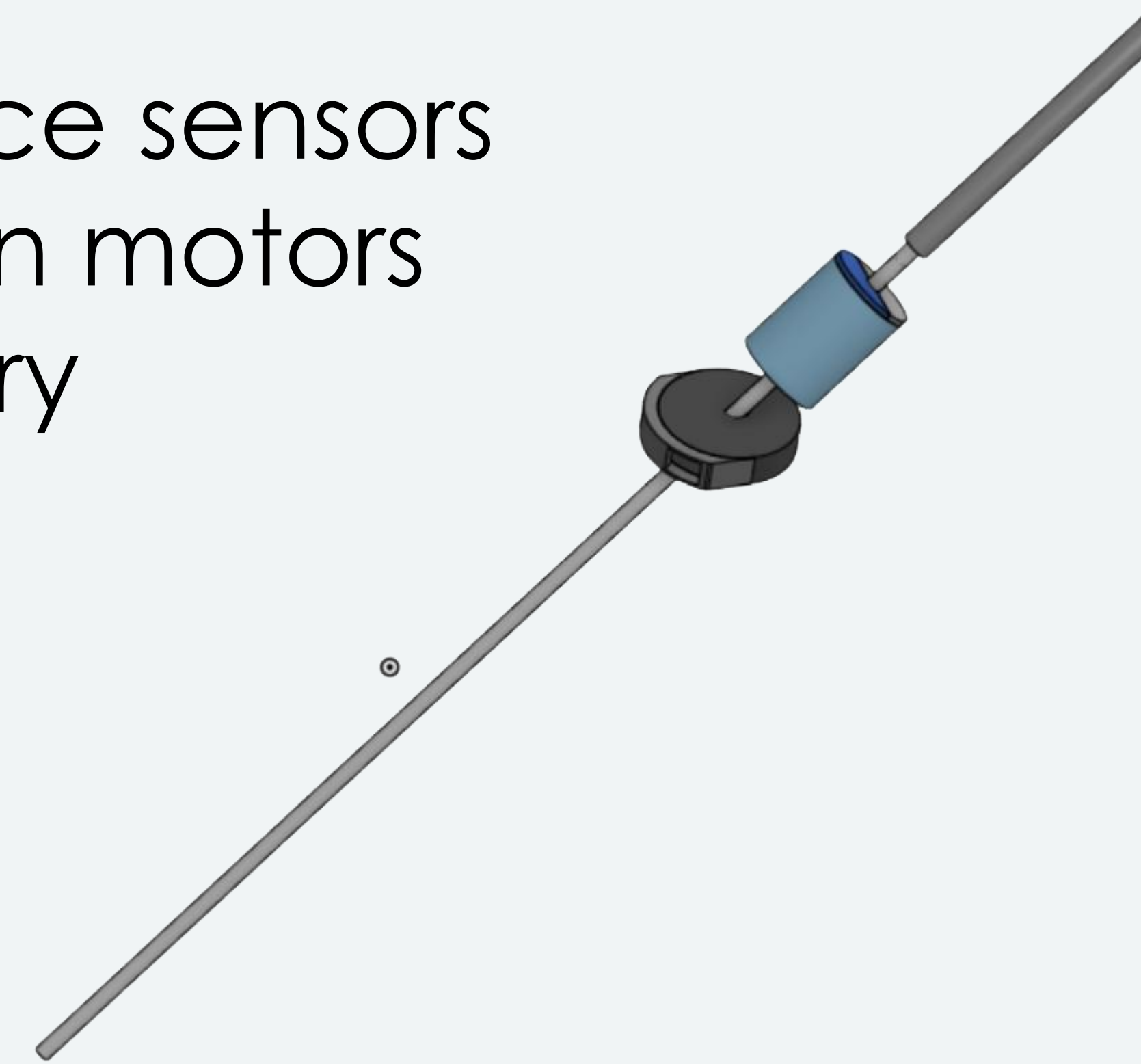
THE SOLUTION

- Pathfinder uses sensors to map out the surroundings
- Uses haptics & audio cues to alert the user
- Detects obstacles beyond the reach of traditional canes
- More affordable than most existing smart canes

THE TECHNOLOGY

Hardware:

- Custom 3D-printed enclosures
- ESP32 Microcontroller
- Time of Flight (ToF) distance sensors
- Variable intensity vibration motors
- Rechargeable LiPo battery
- AI-powered camera
- Speaker module



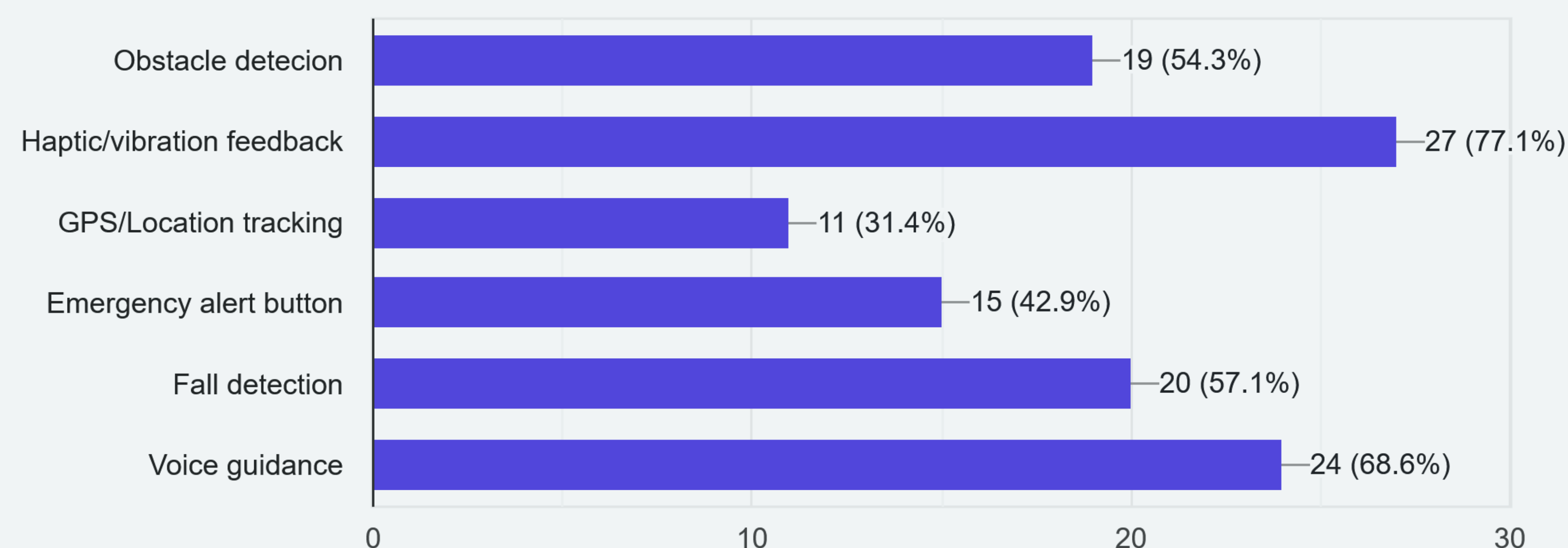
Software:

- Configurable dashboard
- Arduino IDE

VALIDATION

Which features would you find most valuable in a smart white cane?

35 responses



Respondents were willing to pay \$100-\$250 for our product

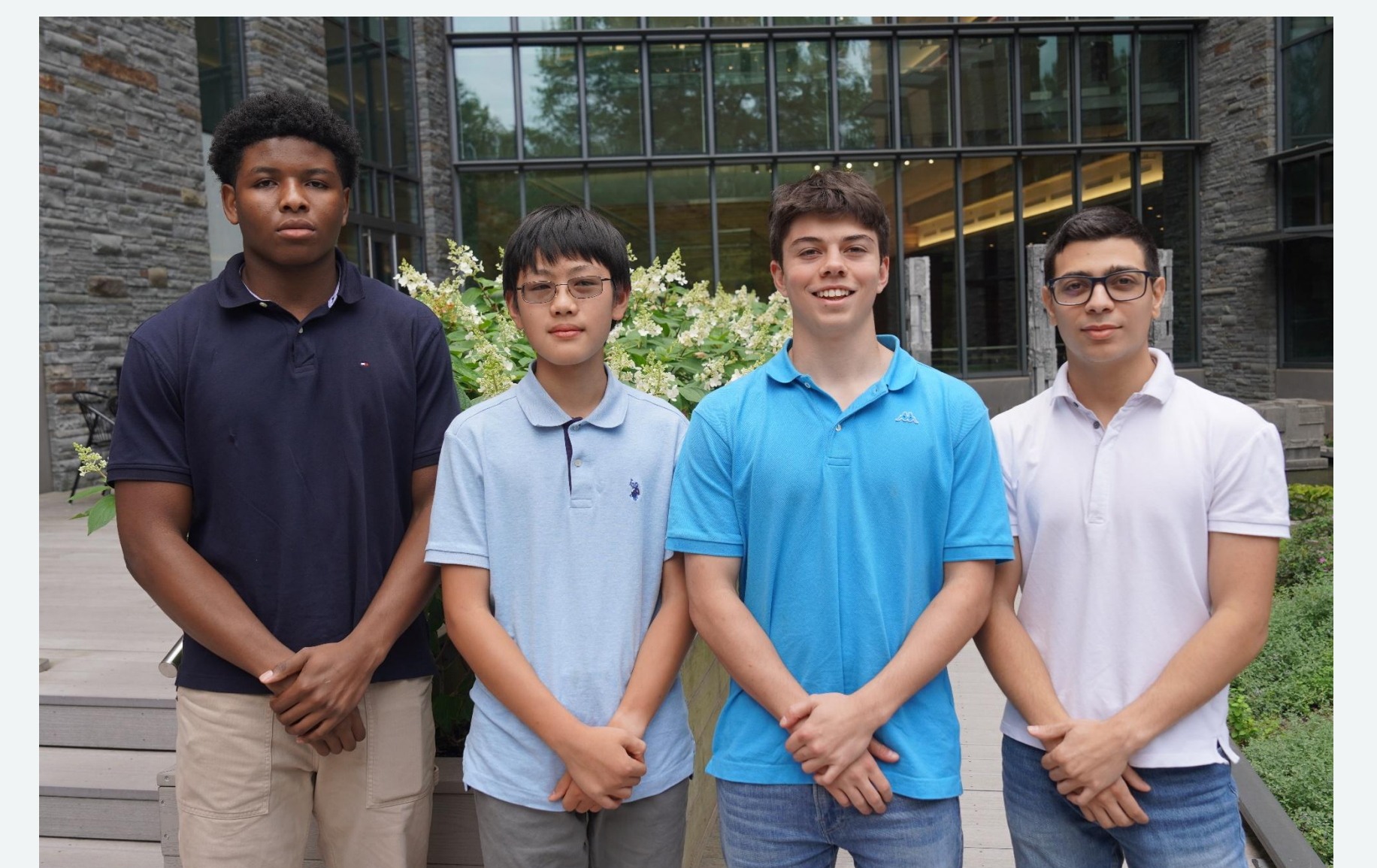
OTHER APPLICATIONS

- Elderly mobility assistance
- Search and rescue in low visibility environments
- Path data storage for daily navigation assistance

FUTURE GOALS

- GPS-based navigation & turn-by-turn guidance
- Fall detection
- Increased accessibility features (language, text-to-speech, etc.)
- A companion app for more feedback and customization

THE TEAM



Artur Khachatryan: Team leader
Bryndon Ekabe: Software Lead
Dominic Ye: Hardware Specialist
Maximo Tode: CAD specialist